

CONSUMER BEHAVIOUR

Student Name: **Aryan Hakeem**

Student ID: **947527**

Student Email Address:

a.hakeem@campus.unimib.it



Research questions

How do customer characteristics and purchasing behavior influence spending and satisfaction in an e-commerce platform?

1. How does total customer spending vary across different membership types?
2. What is the relationship between discount application and customer spending?
3. How is customer satisfaction distributed among customers?
4. How does the number of items purchased relate to total customer spending?
5. Does the time since the last purchase influence customer spending behavior?
6. Are there differences in spending and satisfaction across demographic and geographic groups?

About Data

Dataset: **E-commerce Customer Behaviour Dataset**

(<https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset>)

This dataset provides a detailed view of customer behavior on an e-commerce platform. Each record corresponds to a unique customer and includes information on interactions, transactions, and engagement patterns. It is designed to help analyze customer preferences, satisfaction levels, and shopping trends, enabling businesses to make informed, data-driven decisions to improve the customer experience.

Methodology

The dataset was imported into Power BI Desktop from an Excel source and explored using descriptive data visualization techniques. The dataset required minimal preprocessing, as variables were already structured and formatted appropriately for analysis.

A series of interactive visualizations were designed to address specific research questions related to customer spending, satisfaction, purchasing behaviour, and segmentation. Appropriate chart types were selected based on variable characteristics, including column charts for categorical comparisons, line charts for behavioural trends, and donut charts for proportional analysis.

Interactive slicers were incorporated to enable dynamic filtering by city, membership type, and discount application. This approach supports exploratory data analysis and allows users to examine patterns across different customer segments efficiently.

Insights from the Data

The analysis reveals clear differences in customer behaviour across membership types, with higher-tier members contributing significantly more to total spending and reporting higher satisfaction levels. This suggests that loyalty programs and premium memberships play an important role in increasing customer value and engagement.

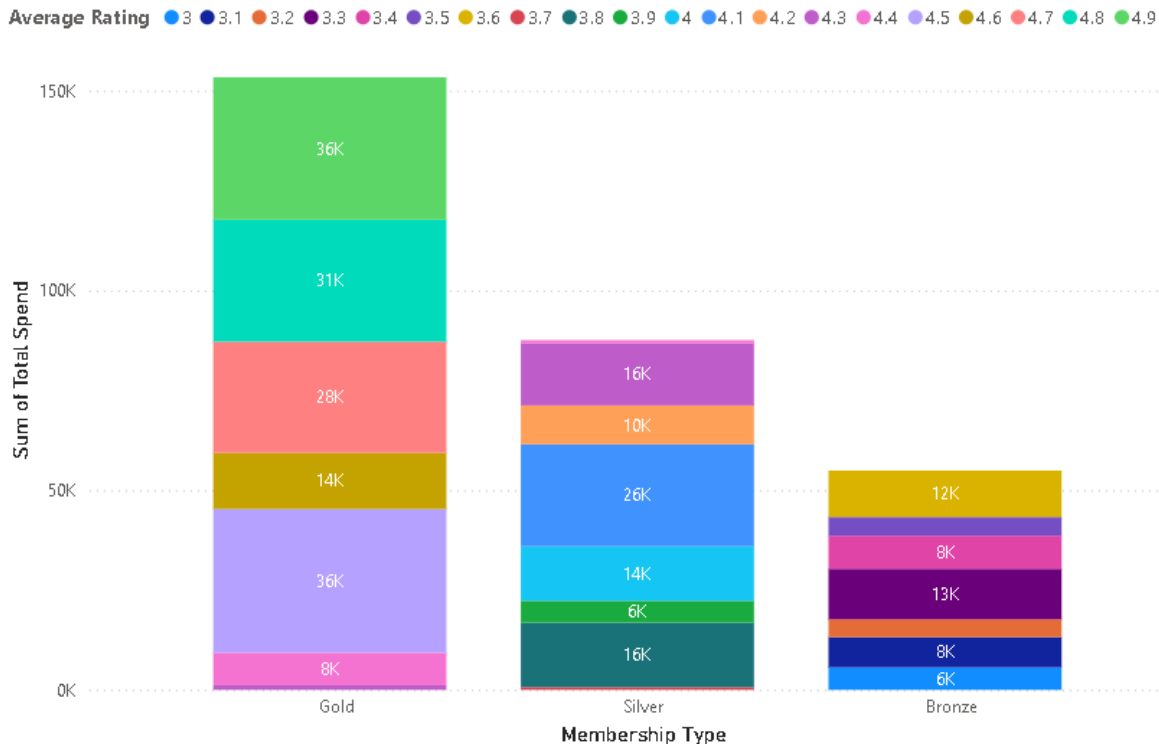
Discount application is associated with higher overall customer spending, indicating that promotional strategies are effective in stimulating purchases, particularly among lower-tier members. Additionally, customers who purchase more items and those who have made recent purchases tend to generate higher total spending, highlighting the importance of basket size growth and retention-focused strategies.

Geographic and demographic variations are also observed in the dataset. Certain cities contribute disproportionately to total revenue, while satisfaction and spending show moderate variation across age and gender groups. Overall, the findings demonstrate that customer behavior in e-commerce is influenced by a combination of demographic characteristics, purchasing patterns, and engagement mechanisms.

Finding: Membership Tier Influences Both Spending and Satisfaction

- This dual-axis chart illustrates the relationship between membership tier and two key metrics: total spending (bars) and average customer rating (line). Gold members show the highest total spending and slightly higher satisfaction ratings compared to Silver and Bronze members, while Bronze members have the lowest spending and ratings. The visualization highlights that **higher-tier membership is associated with both increased revenue and greater customer satisfaction**, providing a clear view of how customer characteristics influence spending and satisfaction.

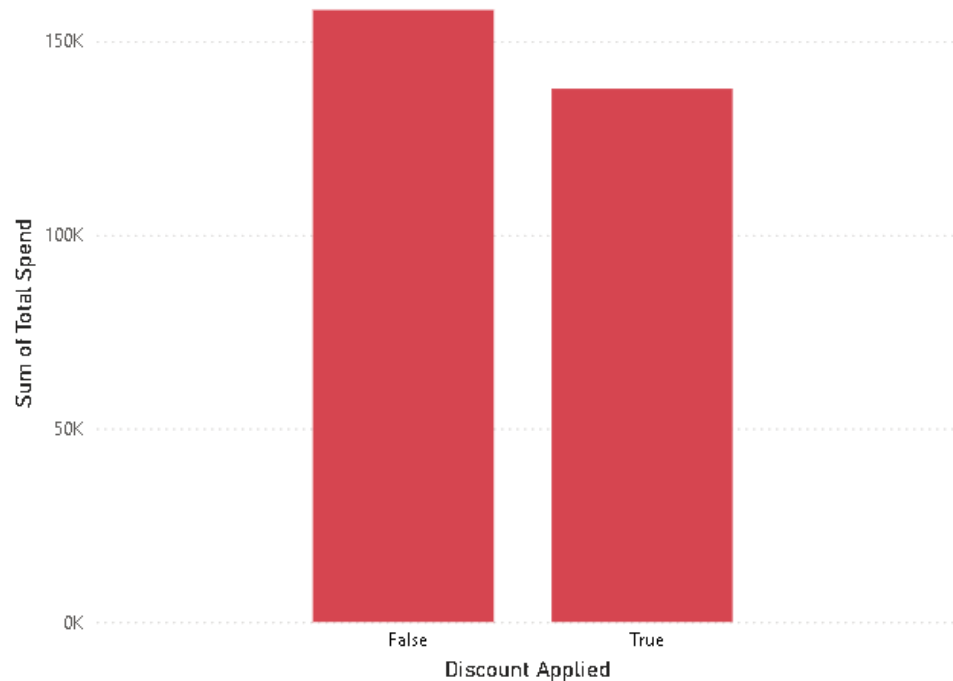
Sum of Total Spend by Membership Type and Average Rating



Effect of Discount Offers on Customer Spending

- The chart compares total spending between customers who received discounts and those who did not. Customers with discounts applied show a higher overall total spend than customers without discounts. This suggests that discount offers are associated with increased purchasing activity and higher spending levels.

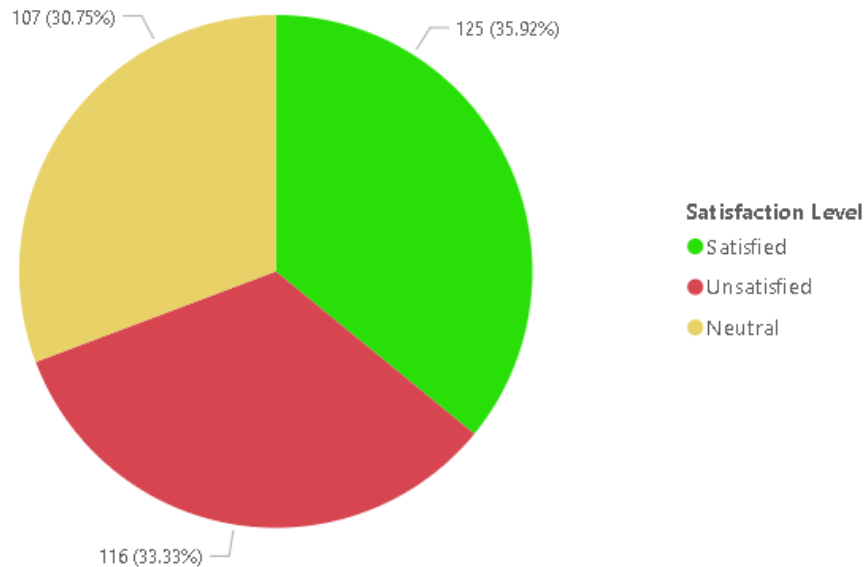
Sum of Total Spend by Discount Applied



Count of Customer ID by Satisfaction Level

- This donut chart illustrates the overall distribution of customer satisfaction. The majority of customers are classified as satisfied, while smaller segments are neutral or unsatisfied. The visualization indicates that customer satisfaction is generally positive, but a significant proportion of customers report dissatisfaction, highlighting potential areas for improvement in the platform's user experience.

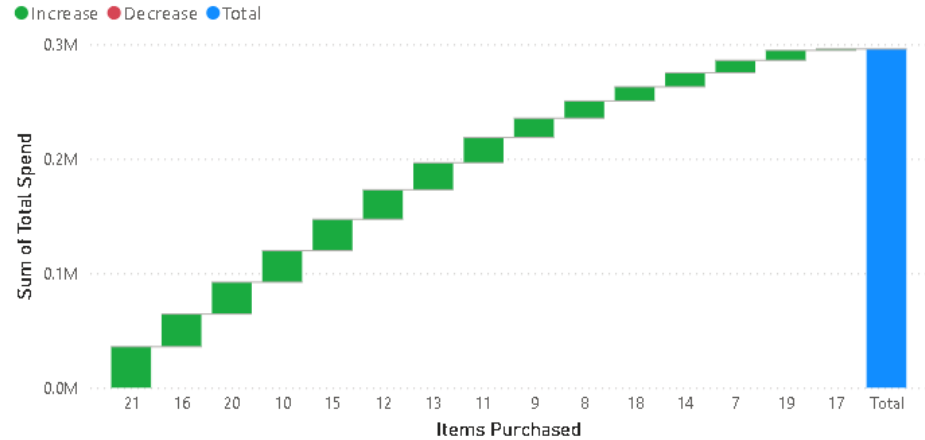
Count of Customer ID by Satisfaction Level



Relationship Between Purchase Volume and Customer Spending

- This chart illustrates the relationship between the number of items purchased and total customer spending. As purchase volume increases, total spending also rises, indicating a positive correlation between the quantity of items bought and spending behaviour. The visualization confirms expected purchasing patterns and highlights the influence of purchase volume on revenue generation.

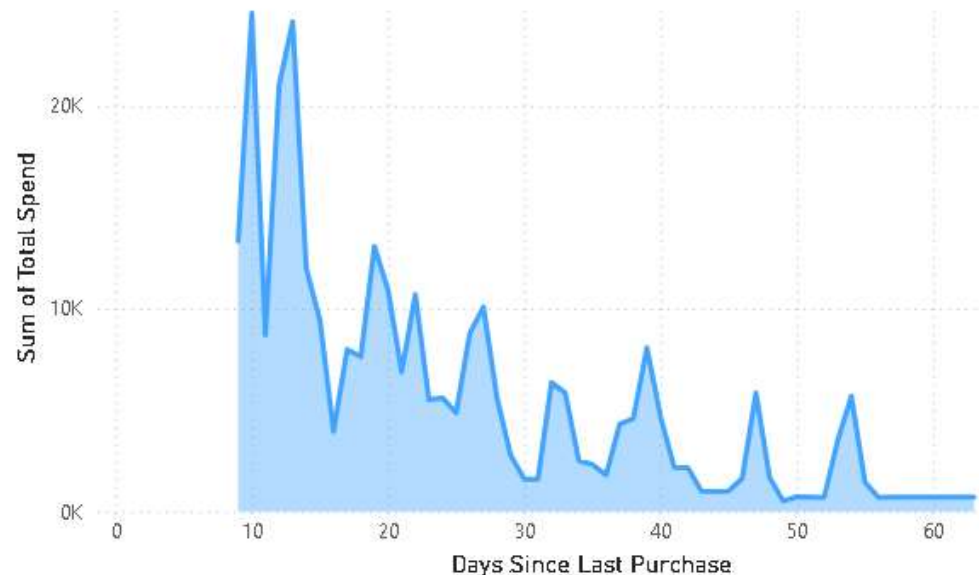
Sum of Total Spend by Items Purchased



Impact of Purchase Recency on Customer Spending

- This line chart illustrates total customer spending in relation to the number of days since the last purchase. Spending is highest among customers who purchased more recently and gradually decreases as the time since the last purchase increases. The visualization highlights the influence of purchase recency on spending behavior, emphasizing the importance of customer retention strategies for driving revenue.

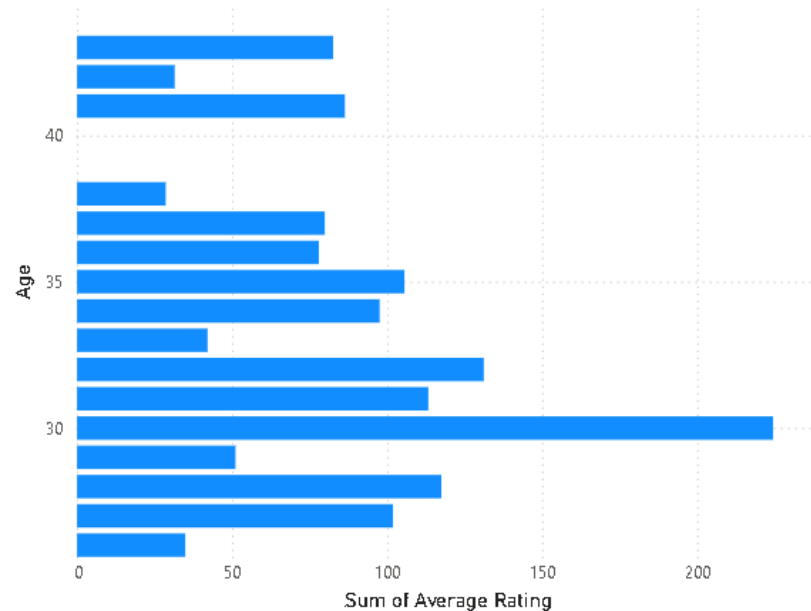
Sum of Total Spend by Days Since Last Purchase



Variation of Customer Satisfaction Across Age Groups

- This chart illustrates average customer ratings across different age groups. Satisfaction levels vary among age ranges, with certain age segments reporting higher ratings than others. The visualization highlights demographic differences in customer satisfaction, providing insights into how age influences user experience on the platform.

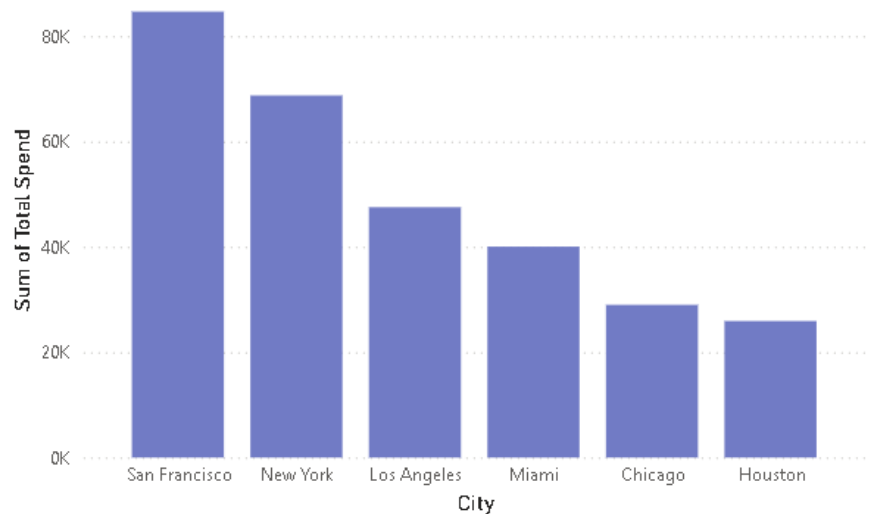
Sum of Average Rating by Age



Geographical Variation in Customer Spending”

- This chart compares total customer spending across different cities. San Francisco and New York exhibit the highest spending, followed by Los Angeles and Miami, while Chicago and Houston show lower spending levels. The visualization highlights geographic differences in customer spending behavior, providing insights into regional trends and opportunities for targeted marketing strategies.

Sum of Total Spend by City



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